

Objective Evidence

****Objective Evidence: NMN****

Mechanisms

- Engineered circ-mRNA exploits exceptional stability and persistent translation capacity inherent to circular RNA scaffolds, driving sustained NAMPT expression and prolonged NAD⁺ elevation (Abstract 1).
- DNA repair-related genomic stress triggers keratinocyte-driven IL-36 production, promoting inflammation in epidermal NAD⁺ deficiency (Abstract 2).
- Modulation of NAD⁺ metabolism (Lu et al., 2026)
- Restoration of As-induced intestinal dysbiosis and enrichment of beneficial genera (Lu et al., 2026)
- Increased Clostridium species, which synthesize vitamin B3 (VB3), a precursor of NAD⁺ (Lu et al., 2026)
- Upregulation of NAD⁺ synthase and NAD⁺ precursor transporter proteins (Lu et al., 2026)
- Reducing inflammation (Abstract 2)
- Encouraging a shift from pro-inflammatory to anti-inflammatory macrophages (Abstract 2)
- Providing strong antibacterial properties (Abstract 2)
- Regulating HAS and NF-κB pathways (Abstract 4)
- Reducing wrinkle formation via the TGF-βR1/Smad3/Collagen pathway (Abstract 4)

Human Data

- Histological integrity and functional preservation confirmed by a single intravitreal circNAMPT-LNP injection matching the neuroprotective efficacy of 14 consecutive daily intraperitoneal NMN doses in dry AMD patients (Abstract 1).
- No direct comparison between human and animal data, but suggests potential as a disease-modifying therapy for dry AMD in humans (Abstract 1).
- Human randomized, double-blind, placebo-controlled trial (Zhao et al., 2026)
- Reduced inflammation, oxidative stress, and wrinkle formation
- Increased skin hydration and barrier function
- Reduced melanin synthesis

Animal Data

- Engineered circ-mRNA exploits exceptional stability and persistent translation capacity inherent to circular RNA scaffolds, driving sustained NAMPT expression and prolonged NAD⁺ elevation (Abstract 1).
- DNA repair-related genomic stress triggers keratinocyte-driven IL-36 production, promoting inflammation in epidermal NAD⁺ deficiency (Abstract 2).
- Animal study using epidermis-specific Nampt conditional knockout (cKO) mice (Abstract 2).
- Animal model (UVB-induced skin damage)
- Animal model (testicular Leydig cells)

Biomarker Effects

- Increased synaptic density in the cortex (Lu et al., 2026)
- Attenuation of autism-like behavioural phenotypes (Lu et al., 2026)
- Reduced inflammatory damage in colonic tissues (Lu et al., 2026)
- Decreased pro-inflammatory cytokine levels in serum (Lu et al., 2026)
- Increased testosterone secretion, cell viability, and antioxidant enzyme activity
- Improved mitochondrial morphology and quantity
- Increased ATP production and mtDNA copy number

Dosage Patterns

- NMN supplementation: 500 mg/L in mice (Lu et al., 2026), 300 mg NMN + 20 mg cornflour in humans (Zhao et al., 2026)
- PQQ supplementation: 20 mg PQQ + 300 mg cornflour in humans (Zhao et al., 2026)
- NMN-modified CeO₂ nanoparticles
- Broccoli extract containing more than 99.0% β-NMN (BRC)

Safety Profile

- No safety findings mentioned in Abstracts 1 and 2.
- No specific safety findings mentioned in Abstract 3.
- The authors declare no competing interests or safety concerns in Abstracts 1, 2, and 4.
- Oral NR and NMN generally well tolerated over weeks to months; short-term safety information available for intravenous NMN (Abstract 3).

Subjective Reports

After analyzing the forum discussions about NMN, I extracted the following information:

****Common Positive Experiences:****

1. ****Improved Energy****: Many users reported increased energy levels and reduced fatigue.
2. ****Enhanced Cognitive Function****: Users experienced improved focus, concentration, and mental clarity.
3. ****Increased Libido****: Some users reported a boost in sex drive and overall libido.
4. ****Reduced Inflammation****: NMN was associated with decreased inflammation and improved joint health.
5. ****Improved Sleep****: Users reported better sleep quality and duration.

****Common Side Effects:****

1. ****Gastrointestinal Issues****: Nausea, diarrhea, and stomach discomfort were common side effects.
2. ****Headaches****: Some users experienced headaches or migraines after taking NMN.
3. ****Dizziness or Lightheadedness****: A few users reported feeling dizzy or lightheaded when standing up quickly.
4. ****Allergic Reactions****: Rare cases of allergic reactions, such as hives or itching, were reported.

****Time to Effect:****

1. ****Initial Effects****: Users typically experienced initial effects within 2-4 weeks of starting NMN supplementation.
2. ****Peak Effects****: Peak benefits were often reported between 6-12 weeks of consistent use.

****Contradictions:****

1. ****Interactions with Medications****: Some users reported interactions with medications, such as blood thinners or diabetes medications.

2. ****Individual Variability****: Users noted that NMN effects can vary significantly from person to person, and some may not experience benefits at all.
3. ****Dosage-Related Effects****: Higher doses of NMN were associated with increased side effects in some users.

****Structured Summary:****

Category	Description
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Positive Experiences	Improved energy, cognitive function, libido, reduced inflammation, improved sleep
Side Effects	Gastrointestinal issues, headaches, dizziness, allergic reactions
Time to Effect	Initial effects within 2-4 weeks, peak benefits between 6-12 weeks
Contradictions	Interactions with medications, individual variability, dosage-related effects

Please note that this summary is based on a general analysis of forum discussions and may not reflect the experiences of every individual who has taken NMN.